AS · Unit 1 · WCH11 Structure, Bonding and Introduction to Organic Chemistry

Formulae, equations and amounts of substance

Atoms are far too small to count one at a time, so chemists count them by weighing, by measuring gas volumes and by measuring solution volumes. This topic builds the idea of the mole, the three equations that link it to things you can measure, and the balanced equations that tell you how much of each substance reacts and forms. Almost every calculation in A level chemistry rests on what is here.

1. The mole and the Avogadro constant

A grain of salt holds about 10^{18} ions, so counting particles directly is hopeless. Instead chemists use a counting unit, the **mole** (symbol mol), in the same way that a baker counts eggs in dozens. One mole of anything contains the same number of particles: the **Avogadro constant**, $L = 6.02 \times 10^{23}\ \text{mol}^{-1}$.

The number was chosen so that the mass of one mole of a substance in grams is numerically the same as its relative mass. One mole of carbon-12 atoms has a mass of exactly 12 g, one mole of water molecules (relative mass 18.0) has a mass of 18.0 g, and so on. That is why weighing is a way of counting.

DEFINITION

Amount of substance is measured in moles. One mole is the amount of substance that contains as many particles as there are atoms in exactly 12 g of carbon-12, which is 6.02×10^{23} particles.

- **Relative atomic mass, A_r:** the weighted mean mass of an atom of an element, measured relative to one twelfth of the mass of an atom of carbon-12. It has no units.
- **Relative molecular mass, M_r:** the mean mass of a molecule relative to one twelfth of the mass of a carbon-12 atom. Add up the A_r values of every atom in the formula. For ionic compounds, which have no molecules, the same sum is called the **relative formula mass**.
- **Molar mass, M:** the mass of one mole of a substance, in g mol^{-1}. It has the same number as M_r: for $CaCO_3$, $M = 40.1 + 12.0 + 3 \times 16.0 = 100.1\ \text{g mol}^{-1}$.
- **Number of particles:** $N = n \times L$, where n is the amount in moles.

Always say which particle you are counting. One mole of water molecules contains two moles of hydrogen atoms, one mole of oxygen atoms and three moles of atoms altogether. One mole of sodium chloride contains one mole of Na^+ ions and one mole of Cl^- ions, so two moles of ions in total.

Counting atoms from a mass

1. How many atoms are there in 5.00 g of aluminium? (A_r: Al = 27.0)
2. Find the amount: $n = \dfrac{m}{M} = \dfrac{5.00}{27.0} = 0.1852\ \text{mol}$

   Keep the unrounded value in your calculator for the next step.
3. Multiply by the Avogadro constant: $N = 0.1852 \times 6.02 \times 10^{23} = 1.11 \times 10^{23}$ atoms

Atoms inside molecules

1. How many atoms are there in 0.500 mol of ammonia, NH_3?
2. Number of molecules: $0.500 \times 6.02 \times 10^{23} = 3.01 \times 10^{23}$
3. Each molecule contains 4 atoms (one N and three H), so the number of atoms is $4 \times 3.01 \times 10^{23} = 1.20 \times 10^{24}$.

The same idea gives the mass of a single atom: one carbon-12 atom has a mass of $12 \div (6.02 \times 10^{23}) = 1.99 \times 10^{-23}$ g.

2. Amount from mass

For any pure solid, liquid or gas, the amount in moles comes from its mass and its molar mass:

$$n = \frac{m}{M} \qquad \text{amount (mol)} = \frac{\text{mass (g)}}{\text{molar mass (g mol}^{-1}\text{)}}$$

The mass must be in grams. Convert first: 1 g = 1000 mg, 1 kg = 1000 g and 1 tonne = 10^6 g. Rearranging gives $m = n \times M$ and $M = m/n$, the last of which is how you identify an unknown substance from a measured mass and amount.

This page uses these relative atomic masses from the periodic table: H 1.0, C 12.0, N 14.0, O 16.0, Na 23.0, Mg 24.3, Al 27.0, S 32.1, Cl 35.5, K 39.1, Ca 40.1, Fe 55.8, Cu 63.5.

EXAM TIP

Give your final answer to the same number of significant figures as the least precise piece of data in the question, usually 3. Never round in the middle of a calculation: carry the full calculator value through and round only at the end, otherwise rounding errors pile up.

Hydrated salts and water of crystallisation

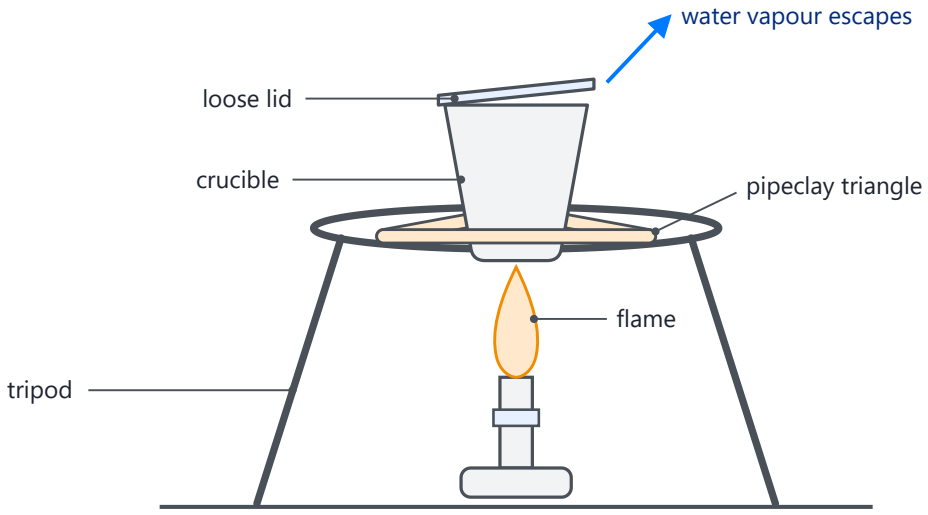
Many salts crystallise from water with water molecules built into their crystal lattice. This is **water of crystallisation**, and the formula shows it after a dot. Blue hydrated copper(II) sulfate is $CuSO_4 \cdot 5H_2O$. Heating drives the water off and leaves white anhydrous $CuSO_4$. The water is part of the compound, so it counts in the molar mass.

Amount of a hydrated salt

1. What amount of $CuSO_4 \cdot 5H_2O$ is in 2.50 g of the blue crystals?
2. $M = 63.5 + 32.1 + (4 \times 16.0) + (5 \times 18.0) = 249.6\ \text{g mol}^{-1}$

   Each H_2O has $M = 18.0$. Forgetting the water, and using 159.6, is the usual error.
3. $n = \dfrac{2.50}{249.6} = 0.0100\ \text{mol}$

To find the number of water molecules, x, in a hydrated salt, heat a weighed sample in a crucible to drive off the water, then weigh the anhydrous solid that is left. The same method measures the mass lost in other thermal decompositions, or the mass gained when a metal such as magnesium is burned to its oxide.



A crucible on a pipeclay triangle. The lid stops solid spitting out, but it sits loosely so that water vapour can escape.

- **Method:** weigh the clean, dry crucible and lid; add about 2 g of the hydrated salt and weigh again; heat strongly for a few minutes; let it cool and reweigh; then reheat, cool and reweigh until two readings agree. This is **heating to constant mass**, and it shows that all the water has gone.
- **Sample size:** about 2 g is right. A much larger sample may not lose all its water. A tiny sample gives a mass change so small that the balance's uncertainty becomes a large percentage of it.
- **Dry crucible:** any water already on the crucible is lost on heating and counted as water of crystallisation, so x comes out too big.
- **Cooling:** cool the anhydrous solid in a desiccator, because many anhydrous salts reabsorb water from the air.

Finding x in $MgSO_4 \cdot xH_2O$

1. 4.92 g of hydrated magnesium sulfate, $MgSO_4 \cdot xH_2O$, is heated to constant mass. 2.40 g of anhydrous $MgSO_4$ is left. Find x.
2. Mass of water lost $= 4.92 - 2.40 = 2.52$ g
3. $n(MgSO_4) = \dfrac{2.40}{120.4} = 0.01993$ mol and $n(H_2O) = \dfrac{2.52}{18.0} = 0.140$ mol

   $M(MgSO_4) = 24.3 + 32.1 + 4 \times 16.0 = 120.4\ \text{g mol}^{-1}$.
4. Divide by the smaller amount: $\dfrac{0.140}{0.01993} = 7.02$, so $x = 7$ and the formula is $MgSO_4 \cdot 7H_2O$.

COMMON MISTAKE

Think about which way an error pushes the answer. If the salt is not heated to constant mass, some water stays behind, the mass lost is too small and x comes out **too low**. If solid spits out of the crucible, the mass lost is too big and x comes out **too high**.

3. Empirical and molecular formulae

- **Empirical formula:** the simplest whole-number ratio of the atoms of each element in a compound.
- **Molecular formula:** the actual number of atoms of each element in one molecule. It is always a whole-number multiple of the empirical formula.

Ethene is C_2H_4 and cyclohexane is C_6H_{12}, but both have the empirical formula CH_2. Analysis of a compound tells you the mass (or percentage by mass) of each element, which only gives the ratio of atoms. You need the relative molecular mass, for example from a mass spectrum, to get the molecular formula.

1. Divide the mass (or percentage) of each element by its A_r. This gives the amount of each element in moles.
2. Divide every answer by the smallest one.
3. If any result is close to x.5, x.33 or x.25, multiply all of them by 2, 3 or 4 to reach whole numbers.
4. For the molecular formula, divide M_r by the empirical formula mass and multiply the empirical formula by the result.

Empirical and molecular formula from percentages

1. A hydrocarbon contains 85.7% carbon and 14.3% hydrogen by mass. Its M_r is 56.0. Find its molecular formula.
2. Take 100 g, so there are 85.7 g of C and 14.3 g of H. Divide by A_r: C: $85.7 \div 12.0 = 7.14$ mol; H: $14.3 \div 1.0 = 14.3$ mol
3. Divide by the smaller value: C: $7.14 \div 7.14 = 1$; H: $14.3 \div 7.14 = 2.00$. The empirical formula is CH_2.
4. Empirical formula mass $= 12.0 + 2.0 = 14.0$, and $56.0 \div 14.0 = 4$. The molecular formula is C_4H_8.

M_r does not need to be exact. The multiple must be a whole number, so a measured value of 55 or 57 still points to 4.

When the ratio is not a whole number

1. An oxide of iron contains 69.9% iron by mass. Find its empirical formula.
2. Oxygen makes up the rest: $100 - 69.9 = 30.1\%$. Fe: $69.9 \div 55.8 = 1.253$; O: $30.1 \div 16.0 = 1.881$
3. Divide by the smaller: Fe = 1, O = $1.881 \div 1.253 = 1.50$
4. 1.50 is not a whole number, so double both: Fe = 2, O = 3. The empirical formula is Fe_2O_3.

Rounding 1.50 to 2 would give FeO_2, which is wrong. Only round values that are within about 0.1 of a whole number.

4. Balanced equations and ionic equations

Atoms are not created or destroyed in a chemical reaction, so a balanced equation has the same number of each kind of atom on both sides. You balance it by changing the numbers in front of formulas, never the formulas themselves. The balancing numbers give the **mole ratio** in which substances react: in $N_2(g) + 3H_2(g) \rightarrow 2NH_3(g)$, 1 mol of nitrogen reacts with 3 mol of hydrogen to make 2 mol of ammonia. State symbols show the physical state: (s) solid, (l) liquid, (g) gas and (aq) dissolved in water.

Ionic equations

When ionic compounds dissolve, the ions separate and move independently. Many of them take no part in the reaction and are present unchanged at the end. These are **spectator ions**. An ionic equation leaves them out and shows only the particles that change, which reveals what is really happening. It must balance for charge as well as for atoms.

- **Split into ions:** soluble ionic compounds in solution (aq), including strong acids such as HCl(aq), H_2SO_4(aq) and HNO_3(aq), and soluble hydroxides such as NaOH(aq).

- **Do not split:** solids (s), including precipitates, insoluble oxides and carbonates; liquids such as water; gases; and covalent molecules.

Writing an ionic equation

1. Start with the balanced full equation, including state symbols:
   $AgNO_3(aq) + NaCl(aq) \rightarrow AgCl(s) + NaNO_3(aq)$

2. Split every aqueous ionic compound into its ions:
   $Ag^+(aq) + NO_3^-(aq) + Na^+(aq) + Cl^-(aq) \rightarrow AgCl(s) + Na^+(aq) + NO_3^-(aq)$

   AgCl is a solid precipitate, so it stays together.

3. Cross out ions that appear unchanged on both sides (Na^+ and NO_3^-):
   $Ag^+(aq) + Cl^-(aq) \rightarrow AgCl(s)$

   Check the charges: 1+ and 1− on the left give zero overall, and so does the neutral solid on the right. Any soluble silver salt mixed with any soluble chloride gives this same equation, which is why ionic equations are so useful.

Displacement reactions

A more reactive metal displaces a less reactive metal from a solution of its compound, because the more reactive metal gives up electrons more readily. Zinc dropped into blue copper(II) sulfate solution becomes coated in red-brown copper, the blue colour fades and the mixture warms up:

$Zn(s) + CuSO_4(aq) \rightarrow ZnSO_4(aq) + Cu(s)$ ionic: $Zn(s) + Cu^{2+}(aq) \rightarrow Zn^{2+}(aq) + Cu(s)$

Halogens behave the same way: a more reactive halogen, which is a stronger oxidising agent, displaces a less reactive one from a solution of its halide ions. The colour change shows the new halogen forming.

Mix	Ionic equation	What you see
chlorine water + potassium bromide	$Cl_2(aq) + 2Br^-(aq) \rightarrow 2Cl^-(aq) + Br_2(aq)$	colourless solution turns yellow-orange
chlorine water + potassium iodide	$Cl_2(aq) + 2I^-(aq) \rightarrow 2Cl^-(aq) + I_2(aq)$	colourless solution turns brown
bromine water + potassium iodide	$Br_2(aq) + 2I^-(aq) \rightarrow 2Br^-(aq) + I_2(aq)$	orange solution turns brown
bromine water + potassium chloride	no reaction	stays orange: bromine is less reactive than chlorine

You will meet the reasons for these trends in the topic on redox and the halogens.

Reactions of acids

An **acid** releases hydrogen ions, $H^+(aq)$, in water. The common strong acids are hydrochloric acid HCl, sulfuric acid H_2SO_4 and nitric acid HNO_3. A **base** neutralises an acid: metal oxides, metal hydroxides, carbonates and ammonia are all bases. An **alkali** is a base that dissolves in water to release hydroxide ions, $OH^-(aq)$, for example NaOH, KOH and aqueous ammonia. A **salt** forms when the hydrogen of an acid is replaced by a metal ion or an ammonium ion.

Reaction	Full equation	Ionic equation	What you see
acid + metal → salt + hydrogen	$Mg(s) + 2HCl(aq) \rightarrow MgCl_2(aq) + H_2(g)$	$Mg(s) + 2H^+(aq) \rightarrow Mg^{2+}(aq) + H_2(g)$	fizzing; the metal dissolves; the mixture warms
acid + alkali → salt + water	$NaOH(aq) + HNO_3(aq) \rightarrow NaNO_3(aq) + H_2O(l)$	$H^+(aq) + OH^-(aq) \rightarrow H_2O(l)$	no visible change; the temperature rises
acid + insoluble base → salt + water	$CuO(s) + H_2SO_4(aq) \rightarrow CuSO_4(aq) + H_2O(l)$	$CuO(s) + 2H^+(aq) \rightarrow Cu^{2+}(aq) + H_2O(l)$	black solid dissolves to give a blue solution
acid + solid carbonate → salt + water + carbon dioxide	$CaCO_3(s) + 2HCl(aq) \rightarrow CaCl_2(aq) + H_2O(l) + CO_2(g)$	$CaCO_3(s) + 2H^+(aq) \rightarrow Ca^{2+}(aq) + H_2O(l) + CO_2(g)$	effervescence; the solid dissolves; the gas turns limewater milky
acid + carbonate solution	$Na_2CO_3(aq) + 2HCl(aq) \rightarrow 2NaCl(aq) + H_2O(l) + CO_2(g)$	$CO_3^{2-}(aq) + 2H^+(aq) \rightarrow H_2O(l) + CO_2(g)$	effervescence
acid + ammonia → ammonium salt	$NH_3(aq) + HCl(aq) \rightarrow NH_4Cl(aq)$	$NH_3(aq) + H^+(aq) \rightarrow NH_4^+(aq)$	no visible change in solution

EXAM TIP

Every neutralisation of a strong acid by an alkali has the same ionic equation, $H^+(aq) + OH^-(aq) \rightarrow H_2O(l)$. In the copper(II) oxide reaction only the sulfate ion is a spectator: the oxide is a solid and the copper ends up in solution, so both change state and must appear.

Precipitation reactions

Mixing two solutions can bring together a cation and an anion that form an insoluble solid. The solid appears as a cloudy **precipitate**, and its colour often identifies the ions.

Ions mixed	Ionic equation	Precipitate
silver and chloride	$Ag^+(aq) + Cl^-(aq) \rightarrow AgCl(s)$	white
silver and bromide	$Ag^+(aq) + Br^-(aq) \rightarrow AgBr(s)$	cream
silver and iodide	$Ag^+(aq) + I^-(aq) \rightarrow AgI(s)$	pale yellow
barium and sulfate	$Ba^{2+}(aq) + SO_4^{2-}(aq) \rightarrow BaSO_4(s)$	white
copper(II) and hydroxide	$Cu^{2+}(aq) + 2OH^-(aq) \rightarrow Cu(OH)_2(s)$	pale blue
iron(III) and hydroxide	$Fe^{3+}(aq) + 3OH^-(aq) \rightarrow Fe(OH)_3(s)$	orange-brown

Making salts

- **Soluble salt from an insoluble base, metal or carbonate:** add the solid to warm acid until no more dissolves, so that it is in excess and all the acid is used up. Filter off the excess solid. Evaporate the filtrate to about half its volume, then leave it to cool and crystallise slowly. Filter off the crystals and dry them between filter papers.

- **Soluble salt from an alkali:** both reactants are solutions, so excess cannot be filtered off. Do a titration with an indicator to find the exact volumes that react, then repeat with those volumes and no indicator, and crystallise as above.
- **Insoluble salt:** mix solutions containing its two ions, filter off the precipitate, wash it with distilled water to remove soluble impurities, and dry it.

The yield of crystals is always below 100%, because some salt stays dissolved in the solution left behind and some is lost on the filter paper and in transfers.

5. Concentration of solutions

A solution is a solute dissolved in a solvent, usually water. Its concentration tells you how much solute is in each cubic decimetre of solution:

$$c = \frac{n}{V} \qquad \text{concentration (mol dm}^{-3}) = \frac{\text{amount (mol)}}{\text{volume of solution (dm}^3)}$$

1 dm^3 is a cube 10 cm on each side, so it equals 1000 cm^3 (one litre), and 1 m^3 = 1000 dm^3. Volumes are usually measured in cm^3, so divide by 1000 before using the equation. Concentration can also be written in g dm^{-3}, the mass of solute in each dm^3: multiply the concentration in mol dm^{-3} by the molar mass.

Concentration of a standard solution

1. 3.18 g of anhydrous sodium carbonate, Na_2CO_3, is dissolved in water and made up to 250.0 cm^3. Find its concentration in mol dm^{-3} and in g dm^{-3}.
2. $M = (2 \times 23.0) + 12.0 + (3 \times 16.0) = 106.0\ \text{g mol}^{-1}$, so $n = \dfrac{3.18}{106.0} = 0.0300\ \text{mol}$
3. Volume in dm^3: $250.0 \div 1000 = 0.2500\ \text{dm}^3$
4. $c = \dfrac{0.0300}{0.2500} = 0.120\ \text{mol dm}^{-3}$
5. In g dm^{-3}: $0.120 \times 106.0 = 12.7\ \text{g dm}^{-3}$

   Check: 3.18 g in a quarter of a cubic decimetre is the same as 12.7 g in a whole one.

Concentrations of ions

A dissolved ionic compound separates into its ions, so the concentration of an ion can be larger than the concentration of the compound. In 0.200 mol dm^{-3} sodium sulfate, $Na_2SO_4(aq) \rightarrow 2Na^+(aq) + SO_4^{2-}(aq)$, so $[Na^+]$ = 0.400 mol dm^{-3} and $[SO_4^{2-}]$ = 0.200 mol dm^{-3}.

Counting ions in a solution

1. How many chloride ions are in 25.0 cm^3 of 0.150 mol dm^{-3} calcium chloride solution?
2. $n(CaCl_2) = c \times V = 0.150 \times 0.0250 = 3.75 \times 10^{-3}\ \text{mol}$
3. Each $CaCl_2$ gives two Cl^- ions: $n(Cl^-) = 2 \times 3.75 \times 10^{-3} = 7.50 \times 10^{-3}\ \text{mol}$
4. $N = 7.50 \times 10^{-3} \times 6.02 \times 10^{23} = 4.52 \times 10^{21}$ ions

Making a standard solution

A standard solution is one whose concentration is known accurately. The aim is to get every bit of the weighed solid into a known final volume.

- **Weigh:** weigh a weighing bottle containing the solid, tip the solid into a beaker, and reweigh the bottle. The difference is the mass transferred. (Alternatively, rinse the weighing bottle into the beaker with distilled water.)
- **Dissolve:** add about 100 cm^3 of distilled water and stir with a glass rod until all the solid has dissolved. Warm gently if needed, and let the solution cool before the next step, because volume changes with temperature.
- **Transfer:** pour the solution through a funnel into a volumetric flask (for example 250 cm^3). Rinse the beaker, rod and funnel with distilled water and add the rinsings to the flask, so that no solute is left behind.
- **Make up to the mark:** add distilled water until the bottom of the meniscus sits on the graduation line, using a dropping pipette for the last few drops, with your eye level with the mark.
- **Mix:** stopper the flask and invert it several times so that the concentration is the same throughout.

Dilution

Adding water to a solution does not change the amount of solute, it only spreads it through a bigger volume. So $n = c_1V_1 = c_2V_2$. The new volume is the original volume plus the volume of water added. To dilute accurately, measure the original solution with a volumetric pipette (it has a much smaller uncertainty than a measuring cylinder) and make it up in a volumetric flask.

Diluting an acid

1. What volume of water must be added to 10.0 cm^3 of 2.00 mol dm^{-3} hydrochloric acid to dilute it to 0.250 mol dm^{-3}?
2. Amount of HCl, which does not change: $n = 2.00 \times 0.0100 = 0.0200 \text{ mol}$
3. Volume needed at the new concentration: $V = \dfrac{n}{c} = \dfrac{0.0200}{0.250} = 0.0800 \text{ dm}^3 = 80.0 \text{ cm}^3$
4. Water to add $= 80.0 - 10.0 = 70.0 \text{ cm}^3$

The concentration fell by a factor of 8, so the final volume must be 8 times the starting volume.

6. Parts per million

Very low concentrations, such as pollutants in water or trace gases in the air, are given in **parts per million (ppm)**, which avoids awkward small numbers.

$$\text{ppm by mass} = \frac{\text{mass of substance}}{\text{total mass of mixture}} \times 10^6 \qquad \text{ppm by volume} = \frac{\text{volume of gas}}{\text{total volume of gas mixture}} \times 10^6$$

1 ppm by mass is 1 mg in every kg. A dilute aqueous solution has a density of about 1 kg dm^{-3}, so 1 ppm is also roughly 1 mg in each dm^3.

Working with ppm

1. A 250 g sample of fish contains 0.100 mg of mercury. What is its mercury content in ppm by mass?
2. Use the same units for both masses: $0.100 \text{ mg} = 1.00 \times 10^{-4} \text{ g}$
3. $\text{ppm} = \dfrac{1.00 \times 10^{-4}}{250} \times 10^6 = 0.400 \text{ ppm}$

4. The other way round: air contains about 420 ppm of carbon dioxide by volume, so 1.00 dm^3 of air holds $420 \times 10^{-6} \times 1.00 = 4.20 \times 10^{-4}\ dm^3$, which is 0.420 cm^3 of CO_2.

7. Volumes of gases

In a gas the molecules are far apart, so the volume a gas takes up depends almost entirely on how many molecules there are, not on how big they are. This is **Avogadro's law**: equal volumes of all gases, at the same temperature and pressure, contain the same number of molecules. It means one mole of any gas occupies the same volume under the same conditions. At room temperature and pressure (r.t.p.) this **molar volume** is taken as 24.0 $dm^3\ mol^{-1}$.

$$n = \frac{V}{24.0} \quad (V \text{ in } dm^3) \qquad \text{or} \qquad n = \frac{V}{24\,000} \quad (V \text{ in } cm^3)$$

Reacting volumes of gases

Because volume is proportional to amount, the balancing numbers in an equation give the ratio of gas volumes directly, as long as all the volumes are measured at the same temperature and pressure. There is no need to convert to moles.

Reacting gas volumes

1. 40 cm^3 of propane is burned in 250 cm^3 of oxygen. All volumes are measured at r.t.p. What is the total volume of gas at the end? $C_3H_8(g) + 5O_2(g) \rightarrow 3CO_2(g) + 4H_2O(l)$
2. Oxygen used: $5 \times 40 = 200\ cm^3$, so $250 - 200 = 50\ cm^3$ of oxygen is left over.
3. Carbon dioxide formed: $3 \times 40 = 120\ cm^3$
4. At r.t.p. the water is a liquid with a negligible volume, so the gas left is $50 + 120 = 170\ cm^3$.

The ideal gas equation

The molar volume of 24.0 dm^3 only applies at room conditions. For any temperature and pressure use the ideal gas equation, which works for any gas or mixture of gases (then n is the total amount of all the gases):

$$pV = nRT$$

Quantity	Unit you must use	Converting
pressure p	Pa	kPa × 1000 = Pa
volume V	m^3	$cm^3 \times 10^{-6} = m^3$; $dm^3 \times 10^{-3} = m^3$
temperature T	K	°C + 273 = K
gas constant R	8.31 $J\ mol^{-1}\ K^{-1}$	given

COMMON MISTAKE

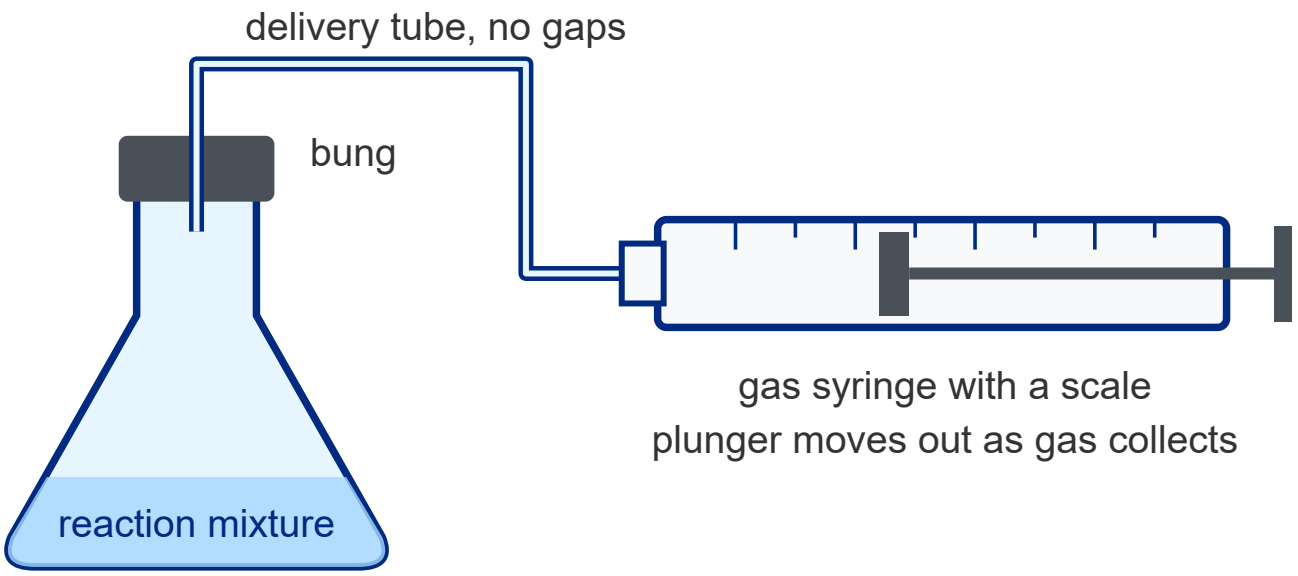
Most wrong answers with $pV = nRT$ come from units. A volume in cm^3 left unconverted is out by a factor of a million, and a temperature in °C gives nonsense. Convert everything to Pa, m^3 and K before you substitute.

Relative molecular mass of a volatile liquid

1. 0.200 g of a volatile liquid is injected into a gas syringe in an oven at 100 °C. It evaporates completely and the vapour occupies 78.0 cm^3 at 101 kPa. Find M_r.
2. Convert the units: $p = 101\,000$ Pa, $V = 78.0 \times 10^{-6}$ m^3, $T = 100 + 273 = 373$ K
3. $n = \dfrac{pV}{RT} = \dfrac{101\,000 \times 78.0 \times 10^{-6}}{8.31 \times 373} = 2.54 \times 10^{-3}$ mol
4. $M = \dfrac{m}{n} = \dfrac{0.200}{2.54 \times 10^{-3}} = 78.7$ g mol^{-1}, so M_r = 78.7. The liquid could be benzene, C_6H_6 (M_r = 78.0).

For a fixed amount of gas, nR is constant, so $\dfrac{p_1V_1}{T_1} = \dfrac{p_2V_2}{T_2}$. This handles a gas moved to new conditions without finding n first. At constant temperature it reduces to $p_1V_1 = p_2V_2$: if 100 cm^3 of gas at 100 kPa is let into an empty 500 cm^3 flask, its pressure falls to $100 \times 100/500 = 20$ kPa.

Measuring a gas volume



Collecting a gas in a gas syringe. The gas volume is read from the scale once the reaction stops; record the room temperature and pressure as well, because the volume depends on both.

A gas syringe lets you measure the volume of gas given off, which can be turned into an amount with $n = V/24.0$ or $pV = nRT$. The same idea is used to measure the molar volume of a gas: react a weighed piece of magnesium ribbon with excess dilute hydrochloric acid and collect the hydrogen. For example, 0.0360 g of Mg ($0.0360/24.3 = 1.48 \times 10^{-3}$ mol) gives the same amount of H_2. If 35.6 cm^3 is collected, the molar volume is $35.6/1.48 \times 10^{-3} = 24\,000$ cm^3 mol^{-1}.

- **Gas lost before the bung goes in:** the volume measured is too small. Keep the reactants apart (for example the solid in a small tube inside the flask) and tip them together after sealing.

- **A sticking plunger:** the gas is not at room pressure when you read the volume.
- **Soluble gases:** carbon dioxide and sulfur dioxide partly dissolve in the water in the flask, so less gas is collected than was made.

8. Moving between quantities

Every quantity you can measure links to amount in moles by one of four routes: mass through the molar mass, number of particles through the Avogadro constant, gas volume through the molar volume (or $pV = nRT$), and solution volume through the concentration. The amount is the hub that connects them all.

KEY IDEA

$$n = \frac{m}{M} \qquad N = nL \qquad n = \frac{V_{\text{gas}}}{24.0\ \text{dm}^3\,\text{mol}^{-1}} \qquad n = cV$$

9. Calculations from equations

Most questions give you a quantity of one substance and ask about another substance in the same reaction. They all follow three steps:

1. Turn the quantity you are given into an amount in moles (mass, gas volume or solution).
2. Use the mole ratio from the balanced equation to find the amount of the substance you want.
3. Turn that amount into the quantity asked for.

Reacting masses and gas volume

1 25.0 g of calcium carbonate is heated until it has fully decomposed. Find the mass of calcium oxide formed and the volume of carbon dioxide at r.t.p.
$CaCO_3(s) \rightarrow CaO(s) + CO_2(g)$

2 Step 1: $n(\text{CaCO}_3) = \dfrac{25.0}{100.1} = 0.2498\ \text{mol}$

3 Step 2: the ratio is 1 : 1 : 1, so $n(\text{CaO}) = n(\text{CO}_2) = 0.2498\ \text{mol}$

4 Step 3: $m(\text{CaO}) = 0.2498 \times 56.1 = 14.0\ \text{g}$ and $V(\text{CO}_2) = 0.2498 \times 24.0 = 5.99\ \text{dm}^3$

Check with conservation of mass: CO_2 has mass $0.2498 \times 44.0 = 11.0$ g, and 14.0 + 11.0 = 25.0 g.

Gas volume at other conditions

1 What volume of hydrogen, at 25 °C and 100 kPa, is made when 0.500 g of magnesium reacts with excess hydrochloric acid?
$Mg(s) + 2HCl(aq) \rightarrow MgCl_2(aq) + H_2(g)$

2 $n(\text{Mg}) = \dfrac{0.500}{24.3} = 0.02058\ \text{mol}$, and the ratio Mg : H_2 is 1 : 1.

3 $V = \dfrac{nRT}{p} = \dfrac{0.02058 \times 8.31 \times 298}{100\,000} = 5.10 \times 10^{-4}\ \text{m}^3$

4 $5.10 \times 10^{-4}\ \text{m}^3 \times 10^6 = 510\ \text{cm}^3$

When one reactant runs out first

If you are given amounts of two reactants, one may be in **excess**. The other, the **limiting reactant**, is used up first and decides how much product forms. Work out how much of each is needed for the other, and see which one runs short.

Finding the limiting reactant

1. 2.00 g of magnesium is added to 50.0 cm^3 of 1.00 mol dm^{-3} hydrochloric acid. What volume of hydrogen forms at r.t.p.?
2. $n(\text{Mg}) = 2.00 \div 24.3 = 0.0823$ mol and $n(\text{HCl}) = 1.00 \times 0.0500 = 0.0500$ mol
3. Mg + 2HCl: 0.0500 mol of HCl reacts with only 0.0250 mol of Mg. There is 0.0823 mol of Mg, so magnesium is in excess and the acid is limiting.
4. $n(\text{H}_2) = \frac{1}{2} \times 0.0500 = 0.0250$ mol, so $V = 0.0250 \times 24.0 = 0.600\ \text{dm}^3 = 600\ \text{cm}^3$

   Using the magnesium would predict 1.98 dm^3, which is impossible: the acid has run out long before.

10. Acid–base titrations

A titration finds the concentration of a solution by reacting it with a solution of known concentration. In an acid–alkali titration a measured volume of one solution goes in a conical flask with a few drops of indicator, and the other is added from a burette until the indicator just changes colour. That point is the **end point**, and it shows that the acid and alkali have exactly neutralised each other.

- **Measure the solution in the flask** with a volumetric pipette (usually 25.0 cm^3), rinsed first with that solution. Rinse the burette with the solution it will hold, so any water left inside does not dilute it.
- **Read the burette** at eye level, at the bottom of the meniscus, to the nearest 0.05 cm^3. The titre is the final reading minus the initial reading.
- **Swirl and go slowly:** stand the flask on a white tile so the colour change is easy to see, and add the solution drop by drop near the end point.
- **Repeat:** the first run is a rough titration to find roughly where the end point is. Repeat accurately until you have **concordant** titres, within 0.10 cm^3 of each other, and use the mean of those.

Indicator	Colour in acid	Colour in alkali	End point (alkali added to acid)
phenolphthalein	colourless	pink	colourless to the first permanent pale pink
methyl orange	red	yellow	red to orange

For a strong acid with a strong alkali either indicator works, because the pH jumps by several units with a single drop at the end point. Use phenolphthalein when the acid is weak, such as ethanoic acid, and methyl orange when the alkali is weak, such as aqueous ammonia. Universal indicator is never used: it changes colour gradually, so there is no sharp end point.

Concentration from a titration

1. 25.0 cm^3 of sodium hydroxide solution is titrated with 0.100 mol dm^{-3} sulfuric acid. The titres are 22.10 (rough), 21.65, 21.55 and 21.60 cm^3. Find the concentration of the sodium hydroxide.
   $H_2SO_4(aq) + 2NaOH(aq) \rightarrow Na_2SO_4(aq) + 2H_2O(l)$

2. Mean of the concordant titres: $(21.65 + 21.55 + 21.60) \div 3 = 21.60\ \text{cm}^3$

   Leave out the rough titre. These three lie within 0.10 cm^3 of each other.

3. Amount of acid used: $n = 0.100 \times \dfrac{21.60}{1000} = 2.160 \times 10^{-3}\ \text{mol}$

4. The ratio H_2SO_4 : NaOH is 1 : 2, so $n(\text{NaOH}) = 4.320 \times 10^{-3}\ \text{mol}$

5. $c(\text{NaOH}) = \dfrac{4.320 \times 10^{-3}}{0.0250} = 0.173\ \text{mol dm}^{-3}$

11. Percentage yield and atom economy

The amount of product calculated from the equation is the **theoretical yield**, the most you could possibly get. In practice you get less:

$$\text{percentage yield} = \frac{\text{actual mass of product}}{\text{theoretical mass of product}} \times 100$$

Yields fall short because the reaction may not go to completion (it may be reversible, or stopped too soon), side reactions make other products, and some product is lost when it is transferred between containers, filtered or purified.

Atom economy asks a different question: of all the atoms you put in, what fraction ends up in the product you want? It comes straight from the balanced equation, so it is fixed for a given reaction however carefully you work.

$$\text{percentage atom economy} = \frac{\text{molar mass of the desired product}}{\text{sum of the molar masses of all the reactants}} \times 100$$

Use the balancing numbers when you add up the molar masses. A process with a high atom economy makes little waste and uses raw materials efficiently, so it is more sustainable. Reactions with only one product, such as additions, have 100% atom economy. When there is a by-product, finding a use for it or selling it improves the economics of the process.

Yield and atom economy for making lime

1. Heating 50.0 g of calcium carbonate produced 25.2 g of calcium oxide. Find the percentage yield and the atom economy. $CaCO_3(s) \rightarrow CaO(s) + CO_2(g)$

2. Theoretical yield: $n = 50.0 \div 100.1 = 0.4995\ \text{mol}$, so the most CaO possible is $0.4995 \times 56.1 = 28.02\ \text{g}$.

3. $\text{yield} = \dfrac{25.2}{28.02} \times 100 = 89.9\%$

4. $\text{atom economy} = \dfrac{56.1}{100.1} \times 100 = 56.0\%$

   Even at 100% yield, 44% of the mass of the limestone leaves as carbon dioxide. Yield measures how well the practical work went; atom economy measures how wasteful the reaction is.

Two ways of making ethanol show the difference. Fermentation, $C_6H_{12}O_6 \rightarrow 2C_2H_5OH + 2CO_2$, has an atom economy of $(2 \times 46.0)/180.0 \times 100 = 51.1\%$. Hydration of ethene, $C_2H_4 + H_2O \rightarrow C_2H_5OH$, has an atom economy of 100%, because every atom ends up in the ethanol.

Quick checks

QUICK CHECK

How many oxygen atoms are there in 0.250 mol of carbon dioxide?

- 1.51×10^{23}
- 3.01×10^{23}
- 6.02×10^{23}
- 4.52×10^{23}

Each CO_2 molecule has two oxygen atoms, so there are $2 \times 0.250 = 0.500$ mol of O atoms, which is $0.500 \times 6.02 \times 10^{23} = 3.01 \times 10^{23}$. The answer 1.51×10^{23} counts molecules, not oxygen atoms.

QUICK CHECK

A compound of nitrogen and oxygen contains 1.40 g of N and 3.20 g of O. Its M_r is 92.0. What is its molecular formula?

- NO_2
- N_2O
- N_2O_4
- NO

N: $1.40/14.0 = 0.100$; O: $3.20/16.0 = 0.200$. The ratio 1 : 2 gives the empirical formula NO_2, with mass 46.0. $92.0/46.0 = 2$, so the molecular formula is N_2O_4. NO_2 is only the empirical formula.

QUICK CHECK

What is the concentration of chloride ions in 0.0500 mol dm^{-3} magnesium chloride solution?

- 0.0250 mol dm^{-3}
- 0.0500 mol dm^{-3}
- 0.150 mol dm^{-3}
- 0.100 mol dm^{-3}

$MgCl_2 \rightarrow Mg^{2+} + 2Cl^-$, so every mole of the salt gives two moles of chloride ions: $2 \times 0.0500 = 0.100$ mol dm^{-3}. 0.150 counts all the ions (Mg^{2+} as well).

QUICK CHECK

What is the ionic equation for the reaction between sodium carbonate solution and hydrochloric acid?

- $CO_3^{2-}(aq) + 2H^+(aq) \rightarrow H_2O(l) + CO_2(g)$ ✓
- $Na_2CO_3(aq) + 2H^+(aq) \rightarrow 2Na^+(aq) + H_2O(l) + CO_2(g)$
- $2Na^+(aq) + 2Cl^-(aq) \rightarrow 2NaCl(aq)$
- $CO_3^{2-}(aq) + H^+(aq) \rightarrow CO_2(g) + OH^-(aq)$

Sodium carbonate is dissolved, so it is written as ions, and Na^+ and Cl^- are spectators. The second option keeps the dissolved sodium carbonate as a formula and leaves spectator Na^+ in the equation.

QUICK CHECK

What volume does 0.100 mol of an ideal gas occupy at 300 K and 100 kPa? (R = 8.31 J mol^{-1} K^{-1})

- 2.49 m^3
- 0.224 dm^3
- 2.49 dm^3 ✓
- 24.9 dm^3

$V = nRT/p = 0.100 \times 8.31 \times 300/100\,000 = 2.49 \times 10^{-3}\ \text{m}^3 = 2.49\ \text{dm}^3$. Leaving the pressure in kPa gives 2.49 m^3, a thousand times too big.

QUICK CHECK

Methyl orange is added to hydrochloric acid in a conical flask, and sodium hydroxide solution is run in from a burette. What colour change marks the end point?

- yellow to orange
- colourless to pink
- red to yellow
- red to orange ✓

Methyl orange is red in acid. The end point is the first orange colour. If the solution has gone yellow you have overshot and added too much alkali. Colourless to pink is phenolphthalein.

QUICK CHECK

Which reaction has an atom economy of 100% for the product shown in bold?

- $CaCO_3 \rightarrow \mathbf{CaO} + CO_2$
- $C_2H_4 + Br_2 \rightarrow \mathbf{C_2H_4Br_2}$
- $Zn + CuSO_4 \rightarrow ZnSO_4 + \mathbf{Cu}$
- $CH_4 + H_2O \rightarrow CO + \mathbf{3H_2}$

An addition reaction has a single product, so every atom of the reactants ends up in it. All the others make a second substance that is not wanted, so their atom economy is below 100% even if the yield is perfect.

QUICK CHECK

A student heats a hydrated salt but does not heat it to constant mass before calculating x, the number of water molecules in the formula. How does this affect the value of x?

- It has no effect, because the mass of the salt is unchanged.
- x is too low, because some water is still in the solid.
- x is too high, because the anhydrous salt seems too heavy.
- x is too high, because extra water is lost.

Water left in the solid makes the "anhydrous" mass too big and the mass of water lost too small. Fewer moles of water per mole of salt means x comes out too low.